THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Assertion (A): Acid should always be added to water and not water to acid. Reason (R): Dilution of acid is a highly exothermic process. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Aluminium is a highly reactive metal, yet it is used to make cooking utensils. Reason (R): Aluminium forms a thin, impervious oxide layer on its surface. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Unsaturated hydrocarbons burn with a sooty flame. Reason (R): They have a higher percentage of carbon so burn incompletely. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Veins have valves but arteries do not. Reason (R): Blood in veins flows at low pressure and must be prevented from flowing backwards. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): Nerve impulse travels from dendrite to axon end in a neuron. Reason (R): Chemical signals cross the synapse in only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): Regeneration is the same as reproduction in all organisms. Reason (R): Planaria grows from a cut piece. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Acquired traits are not inherited by the next generation. Reason (R): Changes in body cells do not change the DNA of germ cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): A concave lens can never form a real image. Reason (R): A concave lens diverges light rays. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A myopic person cannot see distant objects clearly. Reason (R): In myopia the image of a distant object is formed behind the retina. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): Copper wires are used for transmission of electricity. Reason (R): Copper has high resistivity. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): A straight current-carrying wire produces a magnetic field around it. Reason (R): A compass needle placed near the wire gets deflected when the current flows. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Decomposers are called the cleaners of the environment. Reason (R): They break down dead organisms into simple substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Mg burns in oxygen: 24 g Mg gives 40 g MgO. Mass of oxygen used is: 1

(a) 40 g **(b)** 16 g **(c)** 64 g **(d)** 24 g

14. 20 mL of NaOH is neutralised by 10 mL of HCl. To neutralise 50 mL of the same NaOH, the volume of HCl needed is: 1

(a) 25 mL **(b)** 5 mL **(c)** 10 mL **(d)** 100 mL

15. Four metals P, Q, R, S: P displaces Q from its salt; Q displaces R; R displaces S. The least reactive is: 1

(a) P **(b)** R **(c)** Q **(d)** S

16. In the balanced equation $2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$, the coefficient of water is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 7

17. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles. 1
Reason (R): A convex mirror always forms an erect, diminished image and gives a wider field of view.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Myopia is corrected by using a concave lens. 1
Reason (R): A concave lens diverges the rays so that the image is formed farther back, on the retina.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** An ammeter is connected in series in a circuit. 1
Reason (R): An ammeter has a very low resistance.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Two magnetic field lines never intersect each other. 1
Reason (R): At any point in a magnetic field, the field has only one direction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. In a food chain, grass, grasshopper, frog and snake, the energy in the frog is 50 J. Find the energy in the grass and in the snake. 2

22. Why is the mass of the residue less than the mass of lead nitrate taken when it is heated in an open tube? 2

23. How much gypsum (172 g) heated gives how much plaster of Paris? Give the mass of water lost. 2

24. Why are metals in the middle of the reactivity series usually obtained from their oxides rather than sulphides or carbonates? 2

25. An alkane has 22 covalent bonds. Find its molecular formula. 2

26. Why is the pH of the small intestine alkaline and why is that necessary? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Determine the number of C-C and C-H bonds in butane and in isobutane and comment. 3

28. Explain why the human heart muscle needs its own blood supply (coronary arteries). 3

29. A frog's leg muscle twitches when a nerve is stimulated. Explain which nerve and the reflex path. 3

30. Why does cutting a plant stem and keeping it in moist soil give a new plant while cutting a human finger does not give a new human? 3

31. How many chromosomes does each of the following have in humans: sperm, egg, zygote, liver cell? What is the number of autosomes in each? 3

OR

A student wants a 5 times magnified real image on a screen using a convex lens of f = 12 cm. Find the object distance and the screen distance. 3

32. Why does the sky appear whitish near the horizon on a clear day? 3

OR

Three bulbs of 100 W each rated 220 V are connected in parallel with a 220 V source. Find the total current and the equivalent resistance. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Meera had a lunch of chapatti, dal and a glass of milk. The food first entered her mouth, where it was mixed with saliva, and then passed through the oesophagus into the stomach and later into the small intestine. 5

(i) Name the enzyme in saliva that acts on chapatti.

(ii) Name the acid secreted in the stomach.

(iii) Where does the complete digestion of food take place?

(iv) State the role of bile juice in the digestion of fat in milk and explain why the medium in the small intestine must be alkaline.

34. **Case study.** While helping in the kitchen, Ramesh accidentally touched a hot tava and immediately pulled his hand away, even before he realised that it was hot. 5

(i) Name this type of response.

(ii) Name the neuron that carries the impulse from the spinal cord to the muscle of the hand.

(iii) Which part of the nervous system acts as the main centre for this action?

(iv) State two differences between this action and writing a letter.

35. Two pea plants, both tall, produce offspring of which one-fourth are dwarf. A third tall plant crossed with a dwarf plant gives all tall offspring. Deduce the genotype of each of the three tall plants and justify. 5

36. A point source is placed 30 cm from a concave mirror of f = 20 cm. By how much and in which direction should the source move so that the image shifts to infinity? Explain what the beam becomes. 5

OR

An eye of a person has near point at 25 cm and far point at 2 m. Find the power of lens that can bring distant objects into clear focus. Predict the nature of the image formed by the lens of an object at infinity. 5

37. An electric heater has two coils of 20 ohm and 30 ohm that can be used singly, in series or in parallel on a 120 V supply. Find the power in each of the four modes and identify the highest-power mode. 5

OR

Compare the magnetic field of a bar magnet and of a current-carrying solenoid. Explain how a solenoid can be used to lift iron scrap. 5

38. A farmer notices that crops are attacked by insects, so he uses pesticide. Birds that eat insects start to die. Explain the reason with reference to the food chain and propose two sustainable alternatives. 5

OR

Zinc and copper are separately treated with dilute sulphuric acid. State the observations and equations, and explain using reactivity. 5

END OF QUESTION PAPER **Total: 80 Marks**

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